

DOP :
DOS :

NLP PRACTICAL MANUAL - Prof. D V Chandran

EXPT 8 : Implementing chunking using NLTK , SpaCy , Stanza in Python

AIM : WAP to implement chunking using NLTK in Python

THEORY : *Chunking is defined as the process of natural language processing used to identify parts of speech and short phrases present in a given sentence.* There are eight parts of speech namely the noun, verb, adjective, adverb, preposition, conjunction, pronoun, and interjection. Also, in the above definition of chunking, short phrases refer to the phrases formed by including any of these parts of speech.

Chunking can be done to identify and thus group noun phrases or nouns alone, adjectives or adjective phrases, and so on. Chunking is used to get the required phrases from a given sentence. However, POS tagging can be used only to spot the parts of speech that every word of the sentence belongs to.

Application of Chunking : When we have loads of descriptions or modifications around a particular word or the phrase of our interest, *we use chunking to grab the required phrase alone, ignoring the rest around it.* Hence, chunking paves a way to group the required phrases and exclude all the modifiers around them which are not necessary for our analysis. *Summing up, chunking helps us extract the important words alone from lengthy descriptions. Thus, chunking is a step in information extraction.*

Two types of chunking:

- **Chunking up** : Here, have an overview of the information. It just helps us get a brief idea of the given data.
- **Chunking down** : Unlike the previous type of chunking, chunking down helps us get detailed information

ALGORITHM :

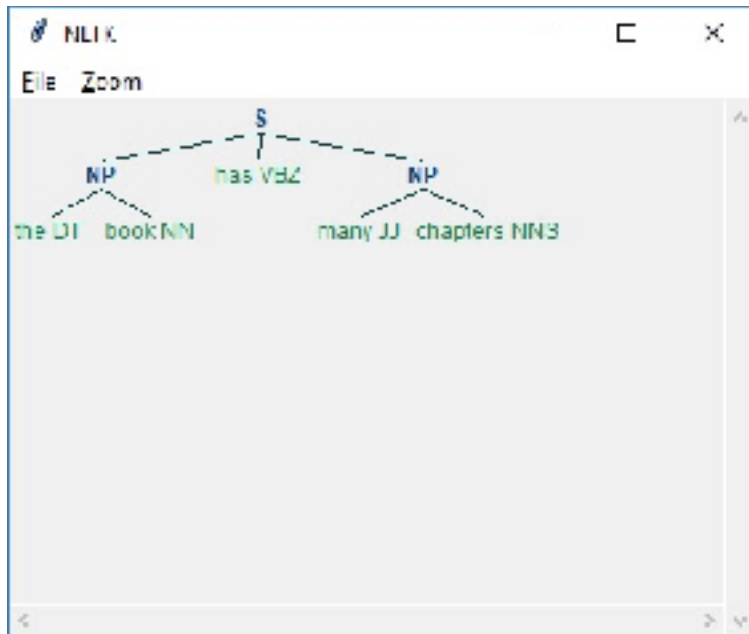
INPUT - OUTPUT :

INPUT : The book has many chapters.

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r ' ' '
NP: {<DT><NN.*><.*>*<NN.*>}
}<VB.*>{
' ' '

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OUTPUT



CONCLUSION :

PROGRAM AND OUTPUT PRINTOUT